5.2 Terminal Velocity Lab Guide

Introduction

As the cart moves along the aluminum track (see figure 1), the magnets on the accessory bracket induce eddy currents in the track. This causes opposing magnetic fields that result in a drag force applied to the cart. The magnitude of this force is measured by analyzing the deceleration of the moving cart. The relationship between the drag force and the speed of the cart is determined.



Figure 5.2.1. Terminal velocity setup and magnetic damping

Theoretical model

Drag force, or fluid resistance, is a force acting opposite to the relative motion of any object moving with respect to its surrounding. Drag force is not a constant force, but depends on the object's velocity:

$$\vec{F}_{drag} = -bv^n\hat{v} \qquad \text{(eq. 1)}$$

Where the minus sign means that the drag force acts in opposite direction to the direction of motion, b is the drag constant that depends on both the material properties and geometry of the objects in relative motion. The drag force is proportional to the velocity for low speed ($n = 1$) and the squared velocity for high-speed ($n = 2$), where the distinction between low and high speed is measured by the Reynolds number.

Let us assume that the magnetic drag force is proportional to the velocity:

$$F_{drag} = -bv \qquad \text{(eq. 2)}$$

Given that a cart of mass m is pushed with initial velocity v_0 on a levelled track, assuming no other sources of friction and no other forces, the cart will decelerate slower and slower since there is only drag force acting in opposite direction to the cart's motion.

To find the functional relation of the cart's velocity on time we write the cart's equation of motion by Newton's 2nd law,

$$ma = -bv \qquad \text{(eq. 3)}$$

The cart's acceleration is the rate of change of the velocity, that is $a = dv/dt$, and equation 3 can be written as a differential equation for the cart's velocity:

$$\frac{dv}{dt} = -\frac{b}{m}v \qquad \text{(eq. 4)}$$

The solution to this differential equation is

$$v(t) = v_0 e^{-\frac{b}{m}t} \qquad \text{(eq. 5)}$$

We get that the velocity will decrease exponentially with a characteristic time $\tau = m/b$.

In the second part of the experiment, we want to find the terminal velocity v_T of a cart released from rest from the top of an inclined plane of angle θ above the horizon (see figure 2).

In this case, two forces act on the cart:

A constant gravitational force down the plane ($mg\sin\theta$) and the drag force up the plane (bv). The cart's equation of motion here is

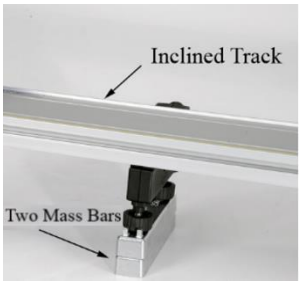


Figure 5.2.2. Inclined track

$$ma = mg\sin\theta - bv \qquad \text{(eq. 6)}$$

As the cart accelerates down the incline, the cart's velocity increase and then also the drag force increases. Eventually, this backwards drag force grows to equal the component of the gravitational force down the plane, and the cart reaches a constant maximal final velocity – the terminal velocity, v_T given is by

$$0 = mg\sin\theta - bv_T$$
$$v_T = \frac{mg\sin\theta}{b} \qquad \text{(eq. 7)}$$

We can also solve the equation of motion (equation 6) to find the cart's velocity at any moment, $v(t)$:

$$m\frac{dv}{dt} = mg\sin\theta - bv \qquad \text{(eq. 8)}$$

The solution to this differential equation is

$$v(t) = \frac{mg\sin\theta}{b}\left(1 - \exp\left(-\frac{b}{m}t\right)\right) \qquad \text{(eq. 9)}$$

Methods

Apparatus

A wireless smart cart is placed on a 1.2 m aluminum track (see fig. 1). The smart cart has internal sensor that can measure the position of the cart as function of time. The cart can be connected via Bluetooth to the computer so the live data can be displayed in the Capstone software.

The magnetic damping accessory can be attached to the end of the cart facing the conductive aluminum track (see fig. 3). When the cart is at motion – the magnetic field generated by the magnets in motion, according to Faraday's Law, will induce eddy currents in the metallic track. According to Lentz's law, the generated eddy currents react to the change in magnetic flux by producing a net force that opposes to the motion of the magnets. This will constitute in what is called magnetic braking and the magnetic friction force will be a drag force!

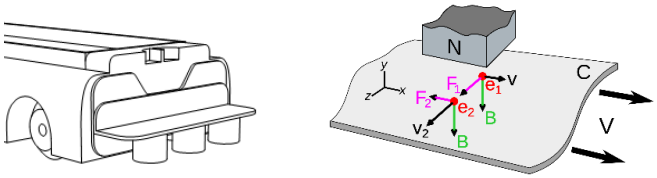


Figure 5.2.3 Magnetic Damping device mounted on the cart [5].

Experimental procedure

Part A: Setting up the system

1. Using the electronic scale weight the mass of: the cart, the magnetic damping device and the two extra masses and record the values in the results and discussion part.
2. Verify that the track is stable and levelled.
3. In PASCO Capstone, click on 'file' and open an experiment named 'TV exp setup.cap' that you have downloaded from Moodle.
4. Turn on the cart and connect to Capstone in the 'hardware setup' button on the top left panel.

Part B: Compensating for the unwanted frictional forces

1. Mount the two extra masses on top of the cart and tape the magnetic damping device on top of the extra masses.
2. Set the cart on the track on the other side from the elastic bumper.
3. Give the cart a small push towards the elastic bumper and click on Record. Your data will look better if you stop recording before the cart hits the bumper. Record the run number and save the data.
4. Using the screw feet (in the opposite side of the elastic bumper), adjust the level of the track so the cart travels at a constant speed towards the elastic bumper.
5. **Record the angle of inclination you have found. Record the run number and save the data.**

Part C: Exploring the drag force

1. Attach the Magnetic Damping Accessory to the cart. The amount of magnetic damping is determined by the distance between the magnets and the track (see Fig. 4). Use the provided spacer to set the distance to about 3 *mm*. Then remove the spacer.
2. In pages: "Part C: F(v)" and "Part C: v(t)" find the graphs of the force and the cart's velocity.
3. **Set the cart on the track on the other side from the elastic bumper and give it a small push towards the elastic bumper and click on Record. Your data will look better if you stop recording before the cart hits the bumper.**
4. **Record the run number and save the data.**

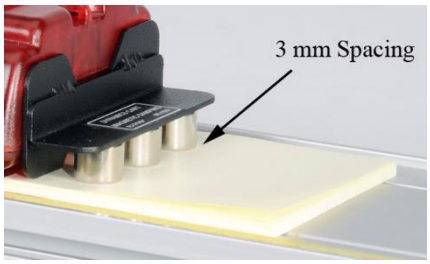


Figure 5.2.3. Spacing for the magnetic damping

Part D: Terminal velocity of the cart on an inclined track

1. Remove the two masses from the cart. Carefully remove the cart from the track without touching the magnetic damping device **(be sure NOT to move the Magnetic Drag Accessory, or you will have to remeasure the coefficient)** and place the cart on the side (so it won't roll from the table).
2. **To increase a slope on the track, place the two masses beneath the track as seen in figure 2.**
3. Use the level meter to find and record the angle of inclinantaion of the track.
4. In page: "Part D: vT" find the graph of the cart's velocity as function of time.
5. **Set the cart on the track on the other side from the elastic bumper, release it from rest and click on Record. Your data will look better if you stop recording before the cart hits the bumper.**
6. **Record the run number and save the data.**

Data analysis

Part A: Setting up the system

1. Open SPYDER and log all the directly measured values (and their uncertainties).
2. Export data from Capstone and import to SPYDER (See the appendix pages 7-8 in the lab guide). **Create a separate data file for each measurement you have performed.**

Part B: Compensating for the unwanted frictional forces

Create two graphs of the $v(t)$ before and after the adjustment of the track level.

Part C: Exploring the drag force

1. Create a graph of the force on the cart and function of its velocity, $F(v)$. Note that you may need to exclude part of the data.
2. **Does the force proportional to the velocity? Use curve fitting to find the proportionally constant.**
3. Create a graph of the cart's velocity as a function of time, $v(t)$. Note that you may need to exclude part of the data (if you pressed record before you pushed the cart, or if you stopped the recording to late).
4. **Does the behavior is as expected? Use curve fitting to find the constant in the exponent and the drag coefficient. Remember the curve fit works best for a linear graph.**
5. **Compare between the 2 values of b you have found. In your opinion, which methods gives the most accurate value?**

Part D: Terminal velocity of the cart on an inclined track

1. Create a graph of the cart's velocity as function of time, $v(t)$. Note that you may need to exclude part of the data (if you pressed record before you pushed the cart, or if you stopped the recording to late).

2. **From this graph find terminal speed v_T of the cart. Compare the experimental value to the theoretical value obtained by eq. 7 and the value of b you have found from the previous part. Compare and discuss your results.**

At the end of work, upload your work files (data file and Python script) to the course's site under the terminal velocity section. Without these files there will not be grade for the lab work!

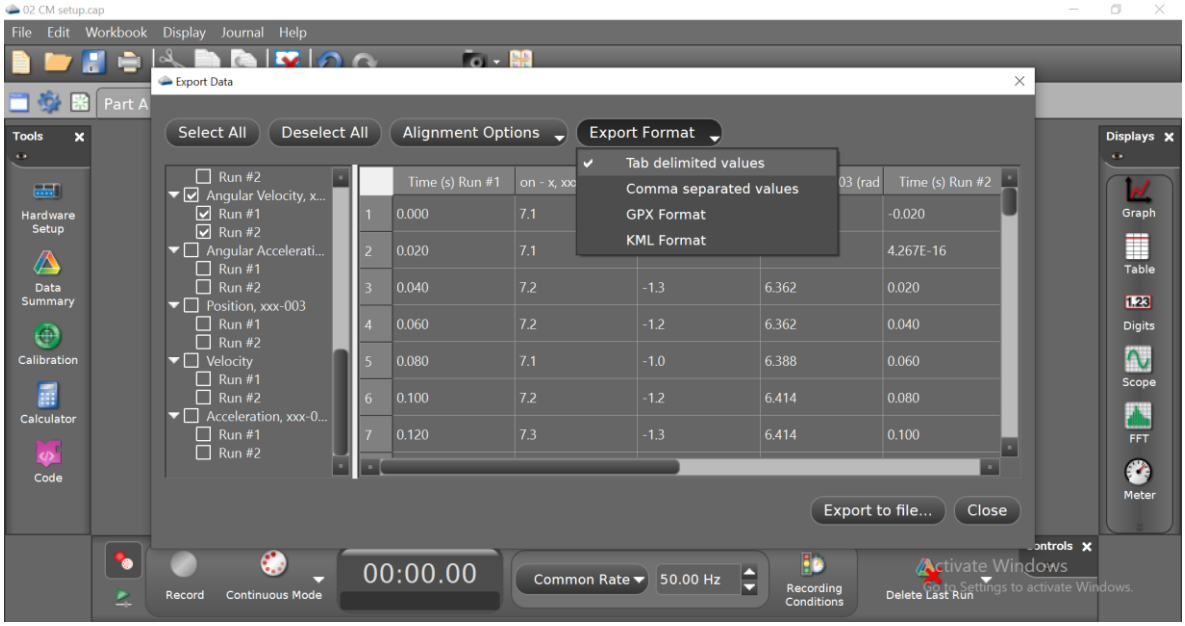
References

[1] Halliday, D., Resnick, R., & Walker, J. (2013). Fundamentals of physics 9th edition. John Wiley & Sons, pp. 121-123.

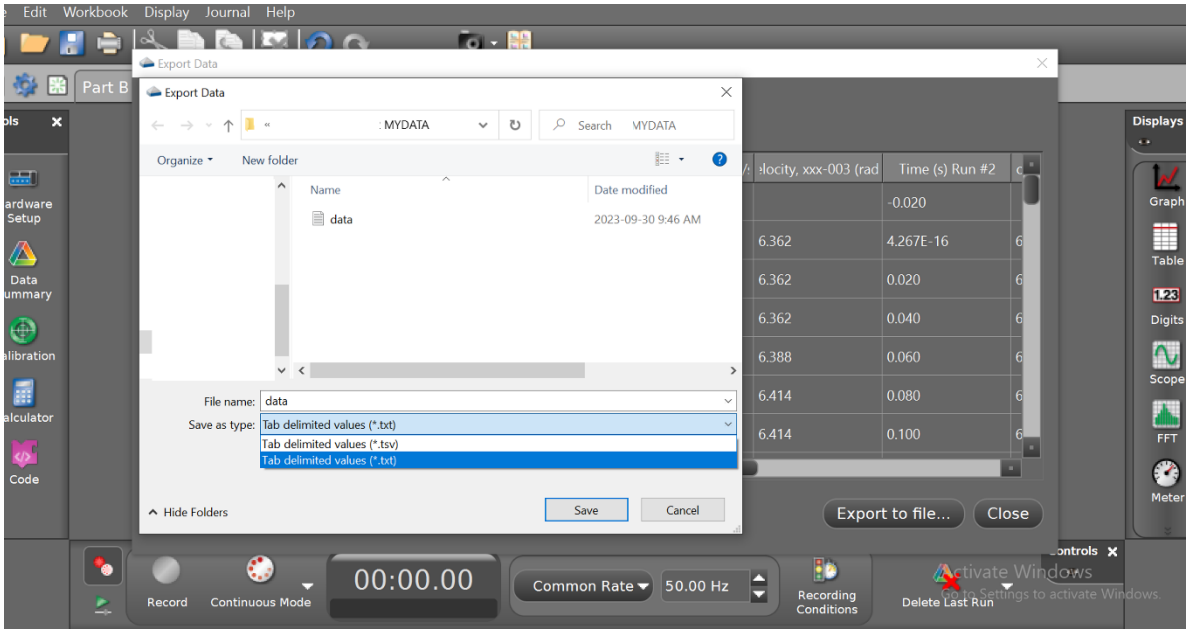
[2] Young, H. D., Freedman, R. A., Sandin, T. R., & Ford, A. L. (2012). University physics 13th edition. Reading, MA: Addison-Wesley. pp. 151-154.

Appendix – How to export data from Capstone and import to SPYDER:

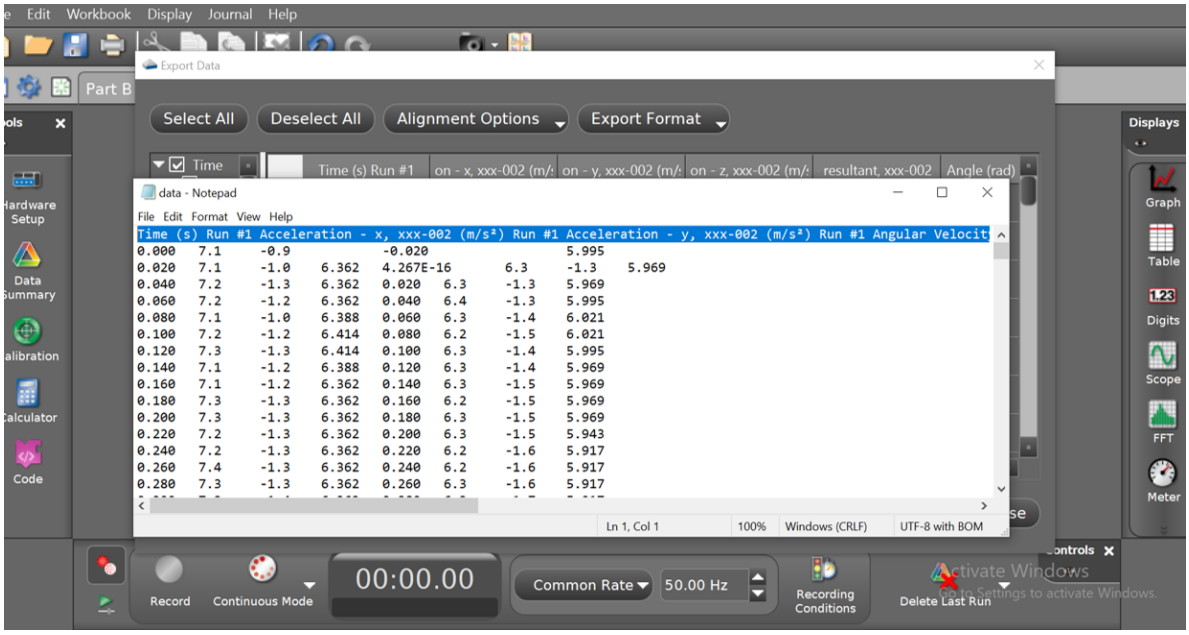
1. In Capstone: Go to File → Export data.
2. In the export data window: Choose Export Format: **Tab delimited values.**



3. On the left ribbon: Leave only the data you want to analyze: time, velocity, force.
4. Go to "**Export to file**" and save as type "**.txt**" in the appropriate folder:



5. Open the data txt file you have created and delete the first line with the text:



6. **Open SPYDER and save the script in the same folder with the data txt file.**
7. Copy the following lines to your script and run:

```
import numpy as np
import matplotlib.pyplot as plt
plt.close('all') #close all open windows in each run
# Set font size 14 for all figures
font = {'size'   : 14}
plt.rc('font', **font)
#Load the experiment data
data = np.loadtxt('data.txt', encoding="utf-8-sig")
```

8. **Create plots of the data to verify that the import worked, and the data is correct.**